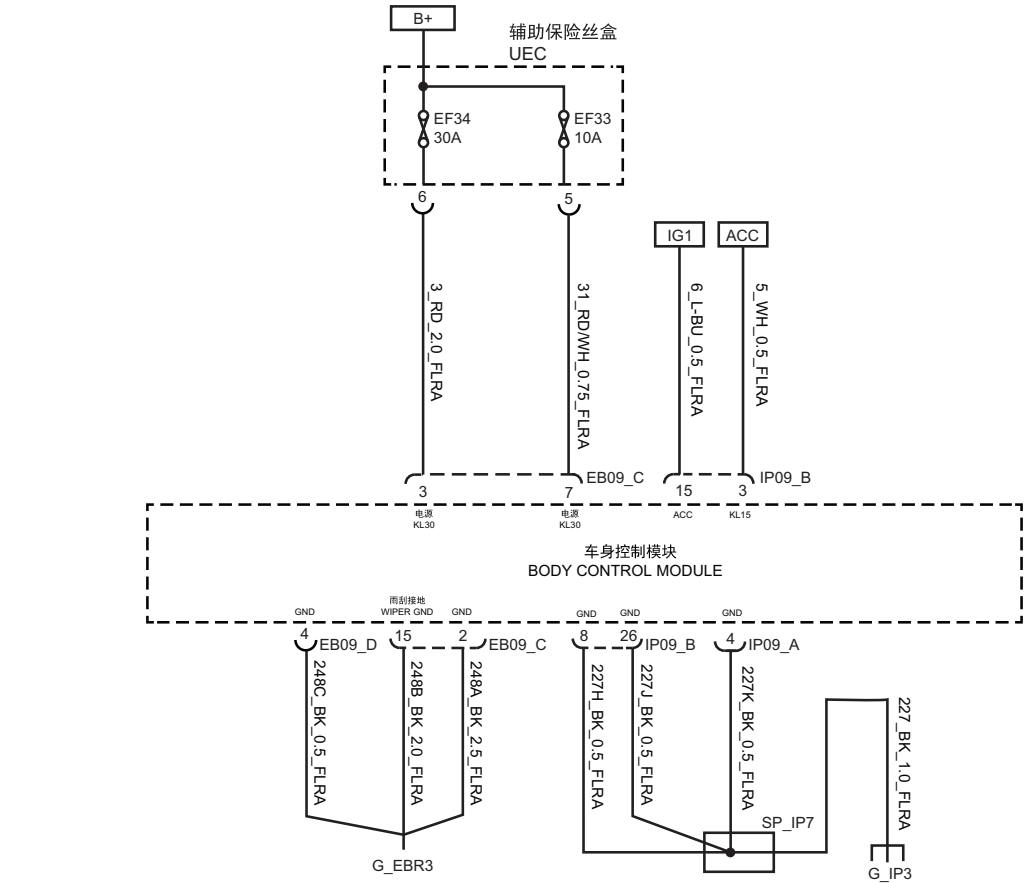
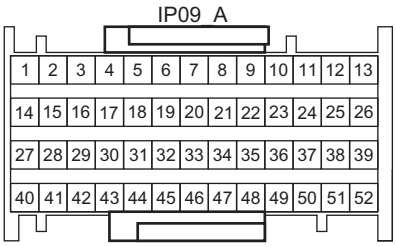
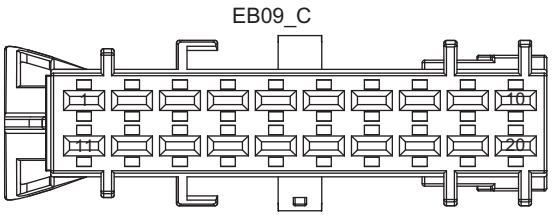
U3003-16、 U3003-17

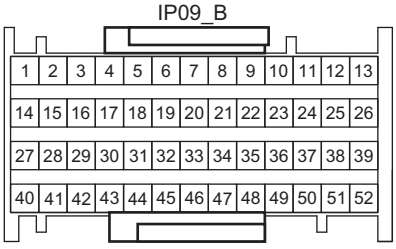
Fault diagnosis code
U3003-16：BCM Voltage below the Low_Threshold
U3003-17：BCM Voltage above the High_Threshold
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Battery • BCM • Charging system • Insurance fuse
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
U3003-16
BCM detects that the battery voltage is greater than 16V, the time is continuously 10s.
U3003-17
BCM detects that the battery voltage is less than 9.6V and the time is continuously 10s.
To set the effect of a fault code condition
1 If a high voltage failure occurs, the internal lamp and the rear window defrost function is closed. 2 If a low voltage failure occurs, the internal light and the rear window defrost function is closed.
Description of circuit diagram
The body control module will continuously monitor whether the sensor and actuator are in normal range. Meanwhile,it will check the connector for virtual connection, short circuit, open circuit and the connection rationality with other signals. In case of any fault, the body control module will save the fault code and enable the safety mode.

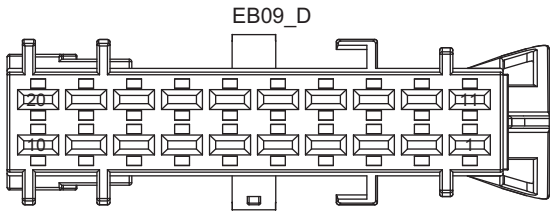
Circuit diagram



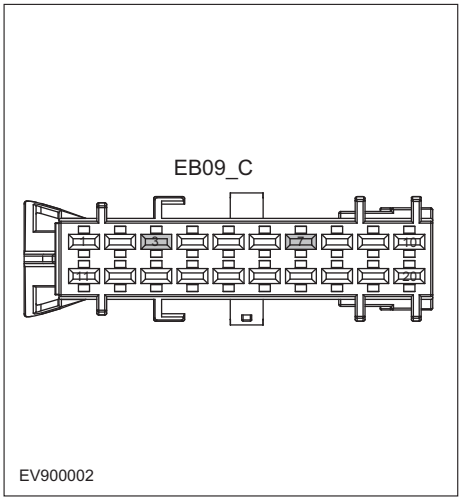


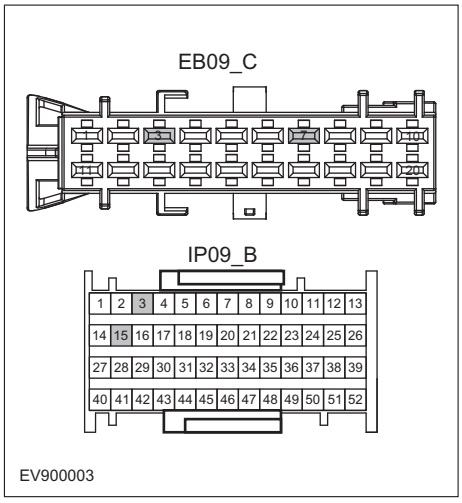




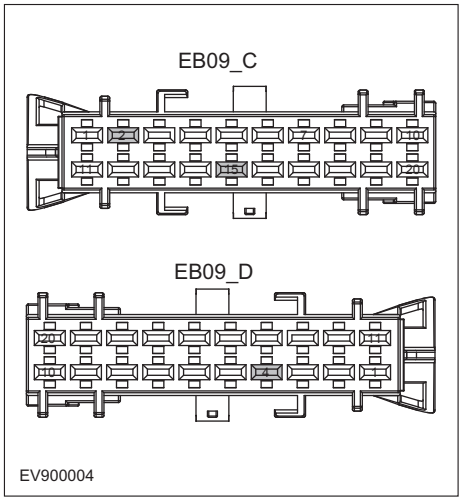
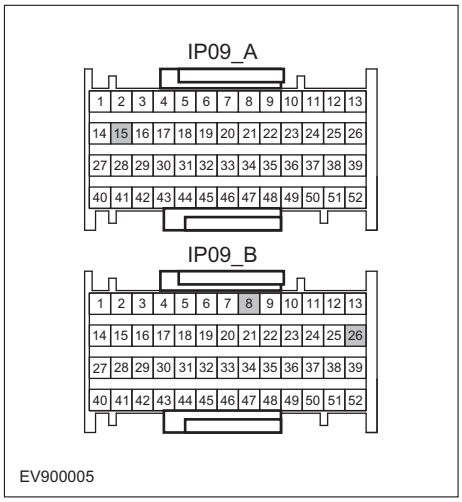


EV900001

Troubleshooting steps		
Step 1	General inspection.	
		1 Check fuses EF33/EF34 for damage. 2 Check battery capacity. 3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Check the body control module with a diagnostic apparatus. 4 If there is any fault code except for U3003-16,U3003-17?
		YES → Refer to:DTC Summary list(BCM).
NO ↓		
Step 3	Check the power voltage of BCM battery.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB09_C, IP09_B of BCM module. 3 Measure the voltage between the harness terminal of BCM and ground. Measuring circuit: voltage between terminal 3 on plug EB09_C and ground. Measuring circuit: voltage between terminal 7 on plug EB09_C and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 4	Check the power voltage when the BCM is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between the harness terminal of BCM and ground. Measuring circuit: voltage between terminal 3 on plug EB09_C and ground. Measuring circuit: voltage between terminal 7 on plug EB09_C and ground. Measuring circuit: voltage between terminal 3 on plug IP09_B and ground. Measuring circuit: voltage between terminal 15 on plug IP09_B and ground. **Standard value:10~14V** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.

YES ↓

Step 5	Test whether the grounding circuit of BCM is open.
 	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB09_D, IP09_A of BCM module. 3 Measure the resistance between the harness terminal of BCM and ground. Measuring circuit: resistance between terminal 2 on plug EB09_C and ground. Measuring circuit: resistance between terminal 15 on plug EB09_C and ground. Measuring circuit: resistance between terminal 4 on plug EB09_D and ground. Measuring circuit: resistance between terminal 15 on plug IP09_A and ground. Measuring circuit: resistance between terminal 8 on plug IP09_B and ground. Measuring circuit: resistance between terminal 26 on plug IP09_B and ground. **Standard value: < 1Ω** 4 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.

<table>
<tr><td colspan="3">YES ↓</td></tr>
<tr><td>Step 6</td><td colspan="2">Test whether BCM is norm.</td></tr>
<tr><td colspan="2"></td><td>1 Turn the ignition switch to OFF.
2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM.
3 Check and repair.
• Corrosion(install new adapting piece or terminal-cleaning module and pin).
• Pin damaged or bent-install new terminal/pin.
• Fly out of pins-install new pins as needed.
4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed.
5 Run the system and determine if the problem persists?</td></tr>
<tr><td colspan="2"></td><td>YES → Replace the BCM.</td></tr>
<tr><td colspan="3">NO ↓</td></tr>
<tr><td colspan="3">At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,</td></tr>
<tr><td colspan="3">Next Step ↓</td></tr>
<tr><td colspan="3">Diagnosis end.</td></tr>
</table>